Amy Nadia Hidayah
3B | 1813331035 | FTE

Selasa, 24 November 2020

MULTIMEDIA (UTS)

① Kode Golomb

(a) n = 19 dan m = 5

$q = \frac{n}{m} = \frac{19}{5} = 3$ → unary code 1110

$r = n - qm = 19 - 3(5) = 4$

$c = \log_2(m) = \log_2(5) = 3$ bit

$s = 2^c - m = 2^3 - 5 = 3$ maka $r > s$

R+S = 4 + 3 = 7 → 111

Code : 1110 111

(b) n = 31 dan m = 15

$q = \frac{n}{m} = \frac{31}{15} = 2$ → unary code 110

$r = n - qm = 31 - 2(15) = 1$

$c = \log_2(m) = \log_2(15) = 4$ bit

$s = 2^c - m = 2^4 - 15 = 1$ maka $r = s$

R+S = 1+1 = 2 → 10

Code 110 10

(c) n = 22 dan m = 8

$q = \frac{n}{m} = \frac{22}{8} = 2$ → unary → 110

$r = n - q(m) = 22 - 2(8) = 6$

$c = \log_2(m) = \log_2(8) = 3$ bit

$s = 2^c - m = 2^3 - 8 = 0$ maka $r > s$

R+S = 6+0 = 6 → 110

code : 110 110

②

Simbol	Probabilitas %
A	50
B	30
C	20

Kode^k untuk simbol Tunstal code 4bit.

$N + K(N-1) \leq 2^n$ → $3 + K(3-1) \leq 2^3$

$2k \leq 5$

$k \leq 2{,}5$

$k \leq 2$

Iterasi 1

Simbol	Probability %
~~A~~	~~50~~
B	30
C	20
AA	2500
AB	1500
AC	1000

Iterasi 2

Simbol	Probability %
~~A~~	~~50~~
B	30
C	20
~~AA~~	~~2500~~
AB	1500
AC	1000
AAA	125000
AAB	75000
AAC	50000

Tunstal code :

4 bit

Simbol	Probability %	Codeword
B	30	0000
C	20	0001
AB	1500	0010
AC	1000	0011
AAA	125000	0100
AAB	75000	0101
AAC	50000	0110

↓ String

↓ Hasil encoding

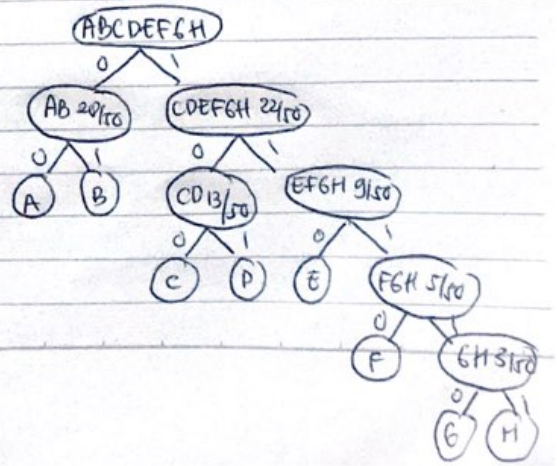
③

Simbol	freq	Simbol	Freq
A	19	E	4
B	9	F	2
C	8	G	2
D	5	H	1

20
22

Tentukan kode-kode setiap simbol

(a) Shannon fano code.



code : A = 00
B = 01
C = 100
D = 111
E = 110
F = 1110
G = 11110
H = 11111

(b) Huffman code.

	Keterangan	
H = 1/50	1	H G F E D C B A
G = 2/50	1	F E D C B A HG^2
F = 2/50	1	D C B A HG^2 FE^2
E = 4/50	1	B A HG^2 FE^2 DC^2
D = 4/50	1	HG^2 FE^2 DC^2 BA^2
C = 8/50	1	DC^2 BA^2 $HGFE^1$
B = 9/50	1	$HGFE^1$ $DCBA^1$
A = 13/50	1	

A B C D E F G H
C D E F G H AB^2
E F G H AB^1 ED^2
G H AB^2 CD^2 EF^2
AB^2 CD^2 EF^2 GH^2
$EFGH^1$ $ABCD^1$
$ABCD^1$ $EFGH^1$

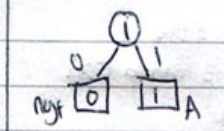
ABCDEFGH 50/50
ABCD

Pohonnya sama seperti Shanon fano

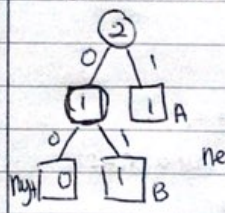
Yang lebih efektif adalah Shanonfanno

(a) DK : Data = ABBCAAB
DH : Metode Adaptive Huffman coding
Jawab

⇒ A → DK = 1 karna $1 \leq k \leq 2^r$ maka :
bit = e+1 = 4+1 = 5bit
biner = K-1 = 1-1 = 0 → (00000)
maka nyt + fixedcode = - 00000

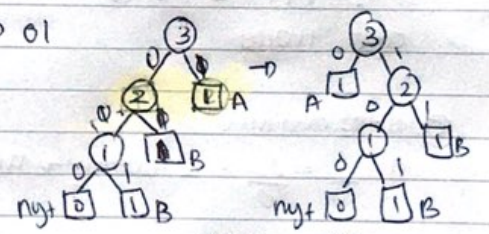


⇒ B → DK = 2 maka $1 \leq K \leq 2^r$ maka
bit = e+1 = 4+1 = 5
biner = K-1 = 2-1 = 1 → 00001
nyt + fixedcode = 0 00001



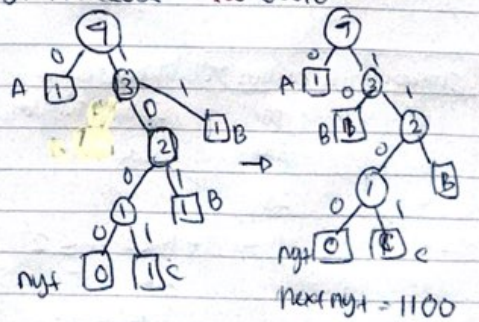
next nyt = 00

⇒ B → 01



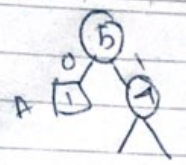
next nyt = 100

⇒ C → DK = 3
bit = e+1 = 4+1 = 5bit
biner = K-1 = 3-1 = 2 → 00010
nyt + fixedcode = 100 00010



next nyt = 1100

> A = 0



⑤

Simbol	Probabilitas %
A	50 = 0,5
B	25 = 0,25
C	25 = 0,25

proses encoding deret simbol CBAB.

a. range setiap simbol

range

A = 0 s/d 0,5

B = 0,5 s/d 0,75

C = 0,75 s/d 1

